



# Affordability Study For Germany

A Machine Learning Perspective: A Retrospect on Urban Landscape Transitions

# Agenda

1. Introduction
2. Data Overview
3. Results
4. Transition
5. Conclusion



# Introduction

- Housing affordability is shifting rapidly across German cities.

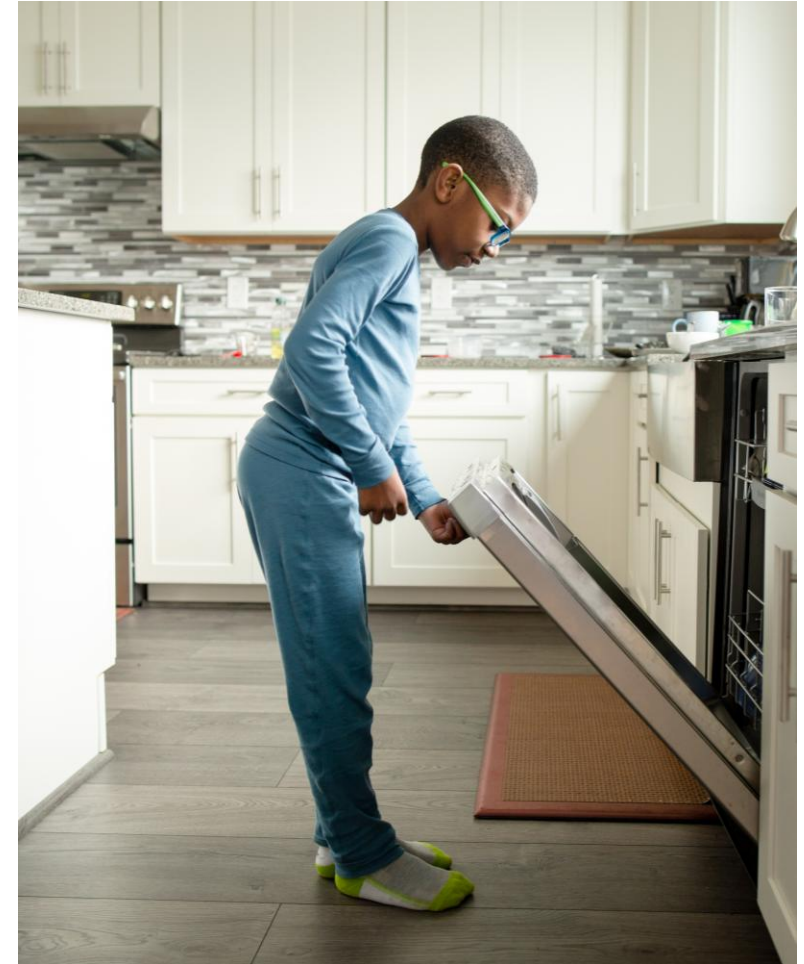
Markets differ not only in price levels, but in:

- supply dynamics
- demographic pressure
- Labour market strength
- investment activity

Goal: Identify natural groups of cities with similar affordability structures.

Method:

- Economic, demographic, Labour, construction & affordability indicators
- Dimensionality reduction (PCA)
- Machine Learning Clustering Algorithm
- SHAP interpretability for feature-driven insights
- Year-over-year and 5-year transition analysis



# Data Sources & Coverage



- Oxford Economics: income, employment, macro indicators
- RIWIS/BFW: housing rents, prices, affordability ratios
- Nexiga: building stock, completions, households, demographics
- Combined: 1694 city-year datapoints, 106 cities, 2010–2024

# Cluster Profiles

|           |                            | Description                                                |
|-----------|----------------------------|------------------------------------------------------------|
| Cluster 0 | High Demand / Large Cities | Structurally expensive, high-liquidity metro hubs          |
| Cluster 1 | Stable & Affordable Cities | Predictable, balanced markets with steady fundamentals     |
| Cluster 2 | Rising Pressure Cities     | Markets under accelerating demand and affordability stress |

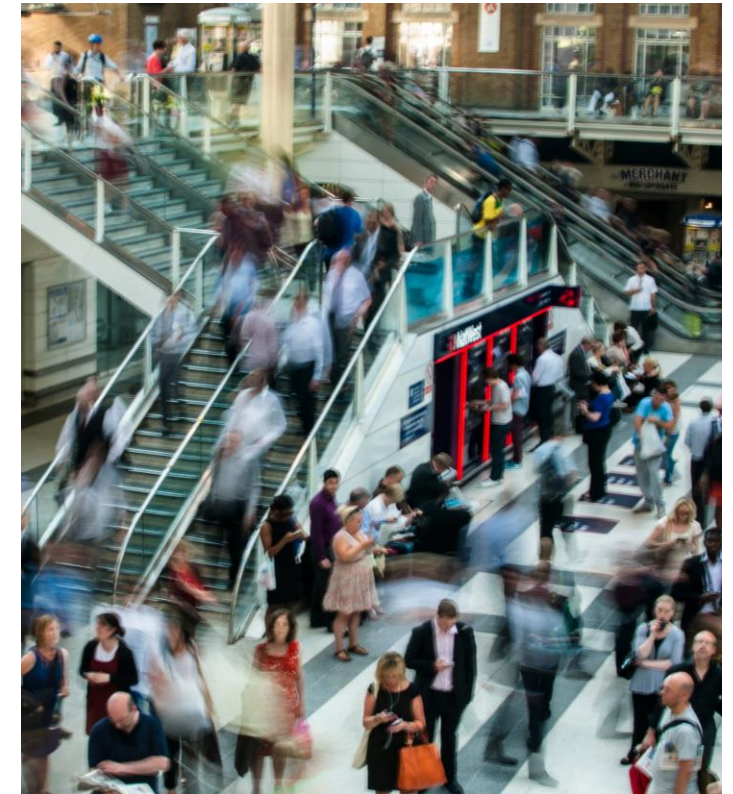
# High Demand / Large Cities

Structurally dominant metropolitan hubs with persistent demand and deep liquidity.

## Key characteristics

- Consistently high housing costs driven by strong economic fundamentals
- Large, diversified labour markets and sustained job creation
- High transaction volumes and strong investor interest
- Tight rental markets supported by migration and student demand
- Very high stability over time, with limited cluster movement

These cities represent low volatility but high entry cost markets. Affordability pressure is structural rather than cyclical, making them resilient but sensitive to regulation and policy intervention.



# Stable & Affordable Cities

Balanced, predictable housing markets forming the backbone of national affordability.

## Key characteristics

- Moderate rents and prices relative to income
- Low to moderate migration pressure
- Balanced supply and demand dynamics
- Lower transaction intensity and limited speculative activity
- Highest stability across the study period

These cities offer defensive, long-term investment environments. They are well suited for stable income strategies, municipal partnerships, and long-hold affordable or workforce housing portfolios.



# Rising Pressure Cities

Growing urban markets experiencing accelerating demand and emerging supply constraints.

- Rapid increases in student population and inward migration
- Rising transaction activity and investor attention
- Supply pipelines lagging demand growth
- Increasing rental and price pressure over time
- Highest frequency of cluster transitions

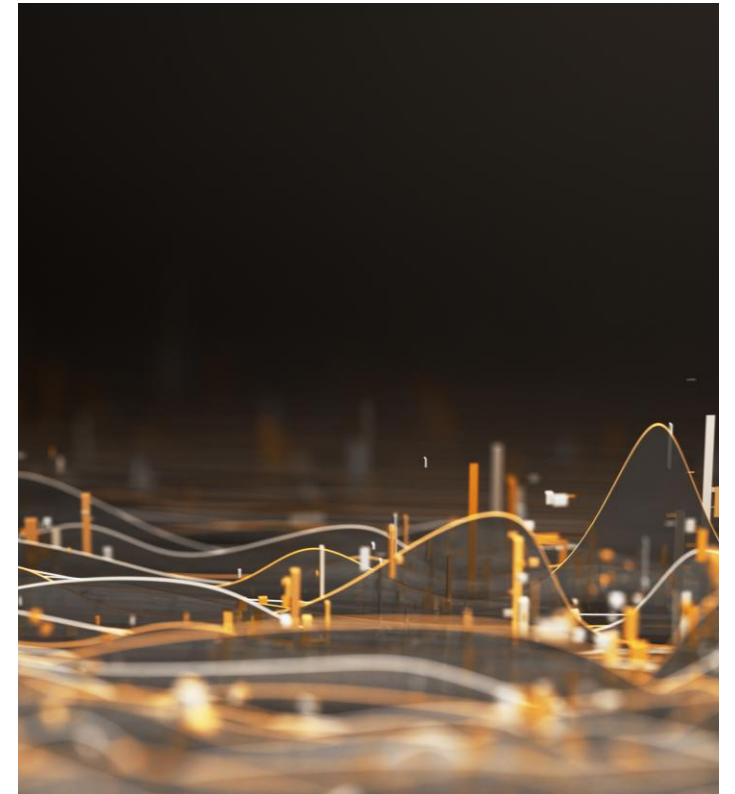
These cities are early-stage stress markets. They present opportunities for forward-looking investment but carry higher affordability risk and growing policy sensitivity if supply does not adjust.



# Underlying Drivers

Top drivers pushing cities into high-pressure clusters:

- Students
- Transactions volume (land & residential)
- Completions vs. permissions mismatch
- Net migration
- Building stock structure (multi-family share)
- GPI total returns
- Office/logistics employment
- GVA sector contributions



# Significance of Top Drivers of Affordability Clusters

| Feature                                             | What It Measures                                  | Why It Matters (Economic Logic)                                                            | Impact on Affordability                                                             |
|-----------------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Students / University Density                       | Number of students enrolled in tertiary education | Students create constant rental demand, low vacancy, intense competition for small units   | Pushes cities into higher-pressure / expensive clusters                             |
| Transactions Volume (Land & Residential Sales)      | Frequency & volume of property trades             | High liquidity signals investor interest, rising valuations, and frequent repricing        | High transaction markets become more expensive                                      |
| Completions vs. Planning Permissions Mismatch       | Gap between approved & completed housing          | Indicates supply bottlenecks (material costs, capacity limits, delays) → demand > supply   | Worsens affordability; clusters drift toward premium/high-pressure                  |
| Net Migration                                       | Inflow minus outflow of residents                 | Migrants create immediate new demand → higher rents, lower vacancy                         | Positive migration = expensive clusters<br>Negative migration = affordable clusters |
| Building Stock Structure (Multi-Family Share)       | Share of multi-unit buildings (BF_WG3M etc.)      | Dense rental stock reflects urban form → higher demand, higher service level, higher rents | High MF share → more expensive; low MF share → more affordable                      |
| GPI(German Property Index) Residential Total Return | Investment return index (MSCI/IPD)                | High returns attract investors, pushing prices and rents upward                            | High GPI = premium, high-pressure clusters                                          |
| Office/Logistics Employment                         | Employment in service & logistics sectors         | Strong job markets pull workers into the city, tightening housing demand                   | More employment → higher demand → higher rents                                      |
| GVA(Gross Value Added) Sector Contributions         | City-level economic output                        | High GVA = strong income base, higher willingness to pay, higher investor inflows          | High GVA regions fall into expensive clusters                                       |

# How Cities Shifted Between Affordability Regions (2010–2025)



## Overall Stability:

79% of German cities remained in the same affordability region across the entire 15-year period.

→ The affordability landscape is structurally stable, not chaotic.



## Cities That Moved to More Expensive Regions

15 cities shifted into a higher-pressure cluster.

This is ~14% of cities showing clear affordability deterioration.

This includes:

Large metros: Düsseldorf, Stuttgart, Cologne, Bremen

High-growth “rising star” cities: Jena, Halle, Nuremberg, Wolfsburg

→ These cities show structural changes in migration, transaction volumes, and student population, matching SHAP drivers.



## Who Stayed Stable?

47 cities remained consistently in the Rising Pressure Cities regime

41 cities remained in the Stable & Affordable Cities regime

Only 3 cities remained in the High Demand / Large Cities regime (Munich, Hamburg, Berlin)  
→ These three behave like structural outliers with consistently unique profiles.



## Cities That Moved to More Affordable Regions

Only 1 city moved downward (Fürth: 2→1 in 2025).

→ Signals rare relief and affordability improvements are the exception.

# Transition Patterns:

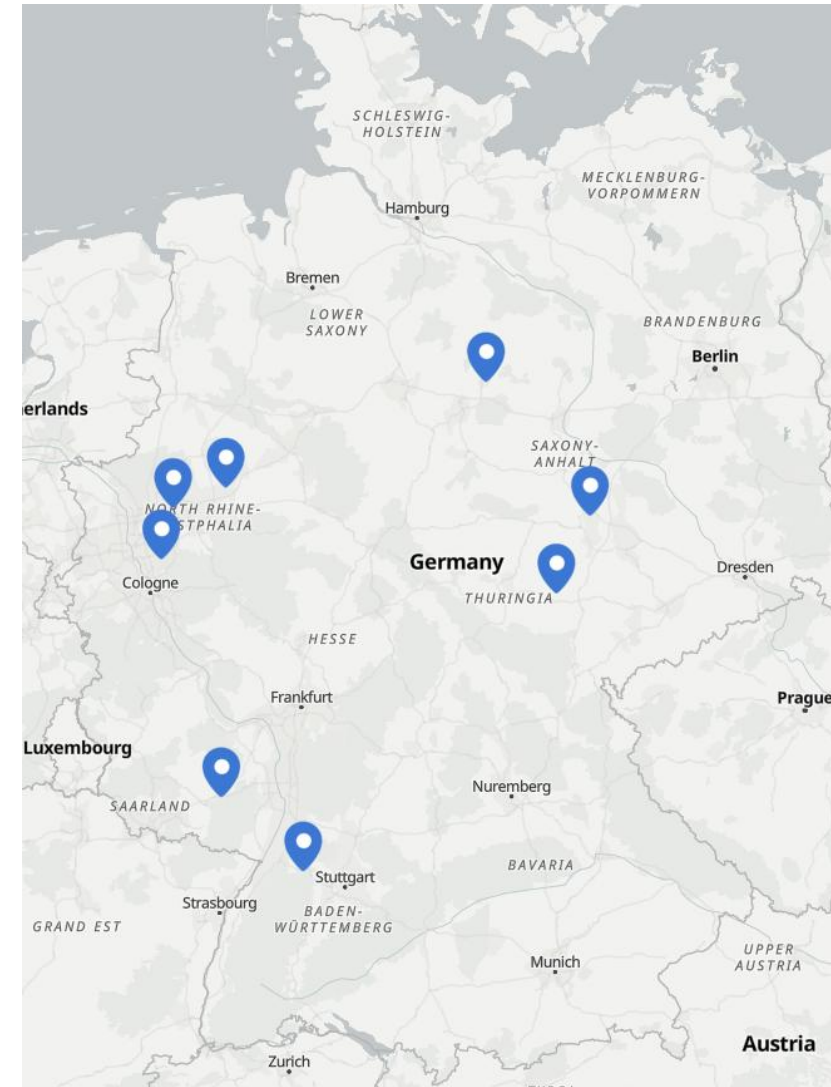
- Cities moving from Stable & Affordable Cities → Rising Pressure Cities

Halle, Hamm, Herne, Jena, Kaiserslautern, Pforzheim, Solingen, Wolfsburg

Why?

- Growing student share
- Increasing migration
- Rising transaction activity
- Supply constraints appearing after 2015

→ *These cities became much hotter markets over 10 years.*



# Transition Patterns:

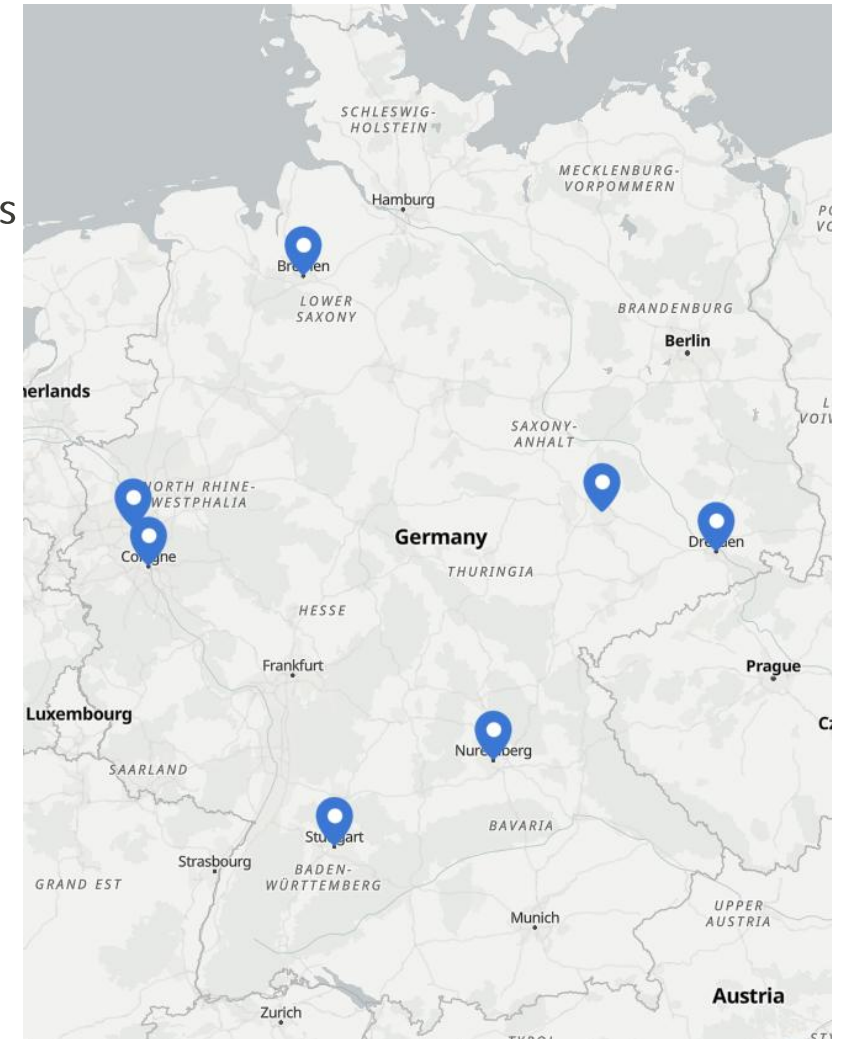
- Cities moving from High Demand / Large Cities → Rising Pressure Cities

Bremen, Cologne, Dresden, Düsseldorf, Nuremberg, Stuttgart, Leipzig

Large cities experiencing pressure intensification beyond historical equilibrium  
**Why?**

- Rapid post-2018 inward migration
- Supply shortages
- Surge in university enrollments
- Transformation of local labour markets

→ *These cities show the strongest structural change , becoming future economic anchors.*



# Transition Patterns:

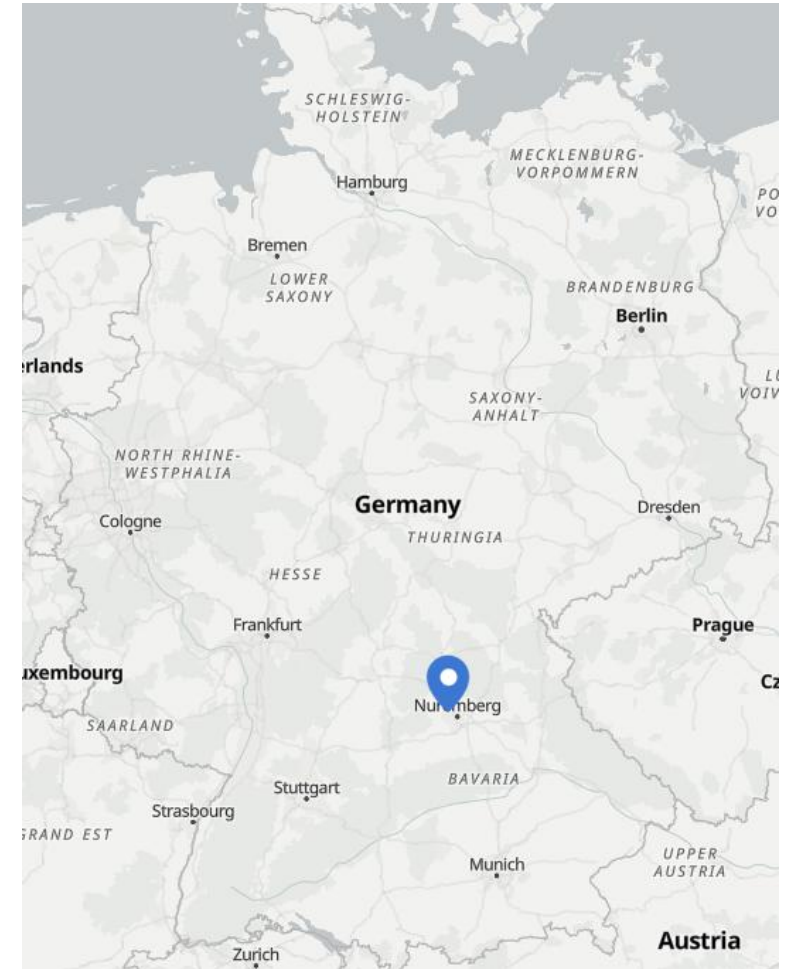
- Cities moving from Rising Pressure Cities → Stable & Affordable Cities

Fürth (unique)

## Why?

- Migration slowed
- Supply increased
- Housing cost pressure stabilized

→ *Only city demonstrating affordability improvement.*



# 5-Year Cycle Insights

- Most changes occur **during economic shocks** (2010→2015, 2020→2025).
- Student population & migration are the earliest signals of transition.
- Transaction activity is the strongest high-pressure trigger.



# Cities Transitioning into Rising Pressure Cities

These are major urban centers that **transitioned into the Rising Pressure Cities regime**

These cities typically exhibit:

- ✓ High net migration
- ✓ Strong student/starter demand
- ✓ Very high sales activity
- ✓ Chronic under-supply
- ✓ Rising GVA & job market depth

→ These cities represent **strategic high-pressure markets** with persistent demand–supply imbalances.

| City      | Affordability Transition                               | Timing / Pattern                                                               |
|-----------|--------------------------------------------------------|--------------------------------------------------------------------------------|
| Bremen    | Rising Pressure Cities<br>→ High Demand / Large Cities | Sustained transition                                                           |
| Dresden   | Rising Pressure Cities<br>→ High Demand / Large Cities | Sustained transition                                                           |
| Nuremberg | Rising Pressure Cities<br>→ High Demand / Large Cities | Sustained transition                                                           |
| Leipzig   | Rising Pressure Cities<br>→ High Demand / Large Cities | Early pressure signals (2010–2013), followed by strong acceleration after 2020 |
| Dortmund  | Rising Pressure Cities<br>→ High Demand / Large Cities | Early pressure signals (2010–2013), intensified post-2020                      |

# Mid-Sized Cities Transitioning out of Stability

Common patterns observed during the transition out of stable affordability:

- ✓ Local migration upticks (Wolfsburg, Bamberg)
- ✓ Strong student presence (Halle, Jena, Kaiserslautern, Pforzheim, Solingen, Fürth)
- ✓ Rising transaction pressure (Pforzheim, Solingen)
- ✓ Labour market tightening (Herne, Hamm)

→ These cities represent **early-stage affordability stress markets**, where pressure is emerging rather than fully priced in.

→ Their **mid-sized profile** makes them suitable for **early monitoring and targeted institutional strategies**.

| City           | Affordability Regime                                |
|----------------|-----------------------------------------------------|
| Halle          | Stable & Affordable Cities → Rising Pressure Cities |
| Hamm           | Stable & Affordable Cities → Rising Pressure Cities |
| Herne          | Stable & Affordable Cities → Rising Pressure Cities |
| Jena           | Stable & Affordable Cities → Rising Pressure Cities |
| Kaiserslautern | Stable & Affordable Cities → Rising Pressure Cities |
| Pforzheim      | Stable & Affordable Cities → Rising Pressure Cities |
| Solingen       | Stable & Affordable Cities → Rising Pressure Cities |
| Wolfsburg      | Stable & Affordable Cities → Rising Pressure Cities |

# Your Priority Markets



## Large Core Metros Moving Up

Berlin didn't move, but these did:

- Cologne
- Düsseldorf
- Stuttgart
- Nuremberg
- Leipzig
- Dresden
- Bremen
- Dortmund

These are becoming high-pressure markets → require rental stabilization, affordability schemes, student housing, and subsidized models.



## Mid-Sized Rising Cities

- Jena
- Halle
- Kaiserslautern
- Pforzheim
- Solingen
- Wolfsburg
- Bamberg
- Mülheim
- Hamm / Herne

These are the sweet spot for:

- Affordable housing investment
- Workforce housing
- Student housing
- Municipal partnerships

# OBSERVATIONS

- 1. No city transitioned into a new High Demand / Large Cities regime.  
High Demand / Large Cities are structural; Rising Pressure Cities emerge over time.

## 2. Most changes occur in 2020→2025 period

12 changes occurred in the last window alone, the strongest structural shift in the whole timeline  
→ matches inflation, migration, housing shortages, and post-COVID dynamics.

- 3. Big cities are moving fastest

This is important for policy makers:  
the pressure is increasing where population is largest.



# Target Cities for Affordable/Social Housing Investment

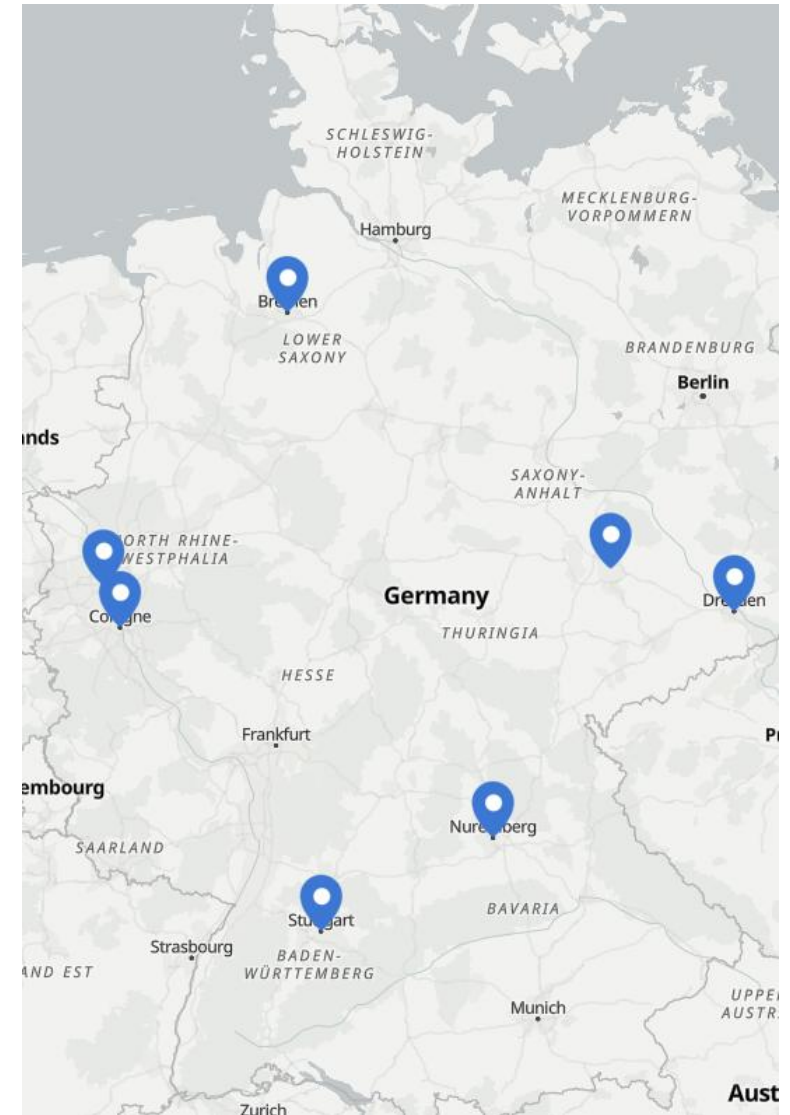
## ■ High-priority opportunity cities (high pressure, undersupplied)

→ Demand strong + supply constraints + student/migration pressure

Leipzig, Cologne, Dresden, Düsseldorf, Stuttgart, Nuremberg, Bremen

These cities moved *into* the “high-pressure” cluster through:

- rising student density (demand floor)
  - rising migration
  - low permissions → completions gap
  - increasing transaction volume → investor demand
  - building stock under pressure (MF share rising)
- These are **strategically relevant markets** for affordable/social housing expansion.



# Stable Affordable Markets (Good for long-term acquisition)

## ■ Cities with consistent stable affordability:

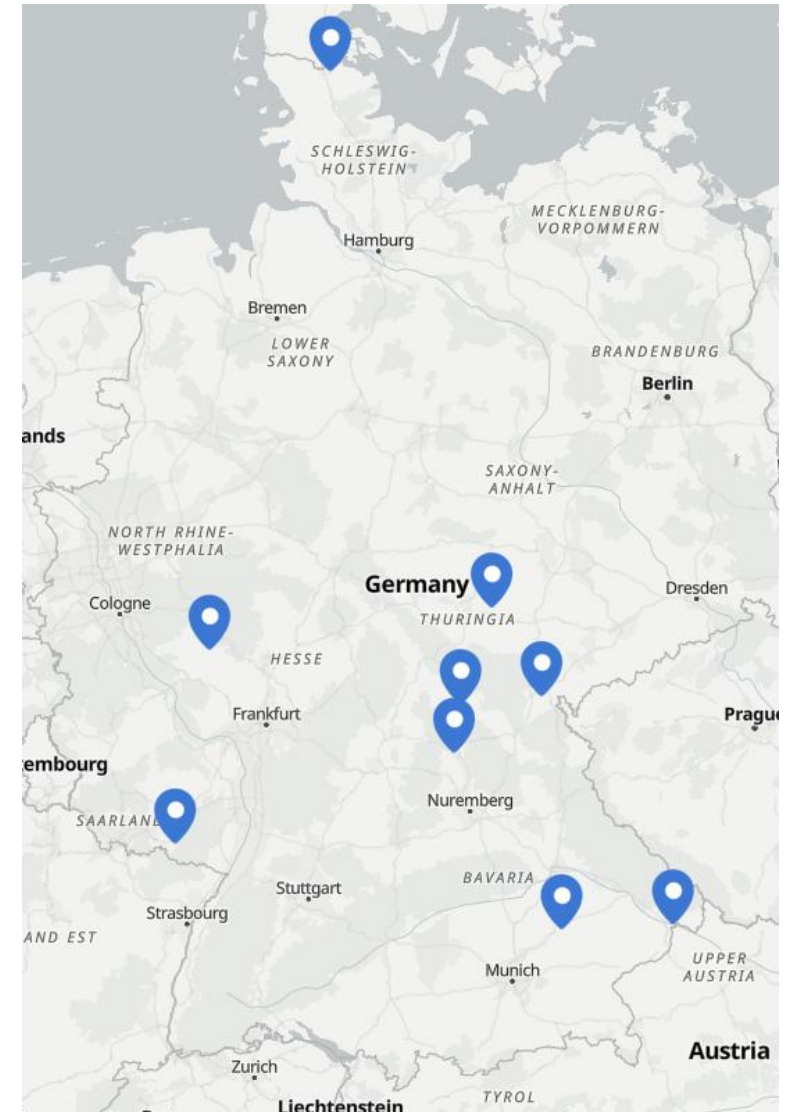
Bamberg, Flensburg, Weimar, Coburg, Hof, Landshut, Passau, Pirmasens

## ■ These offer:

- Predictable rents
- Low migration volatility
- Well-balanced supply & demand
- Lower risk of regulation shock

## ■ Ideal for:

- long-term social housing portfolios
- municipal partnership aligning with stability



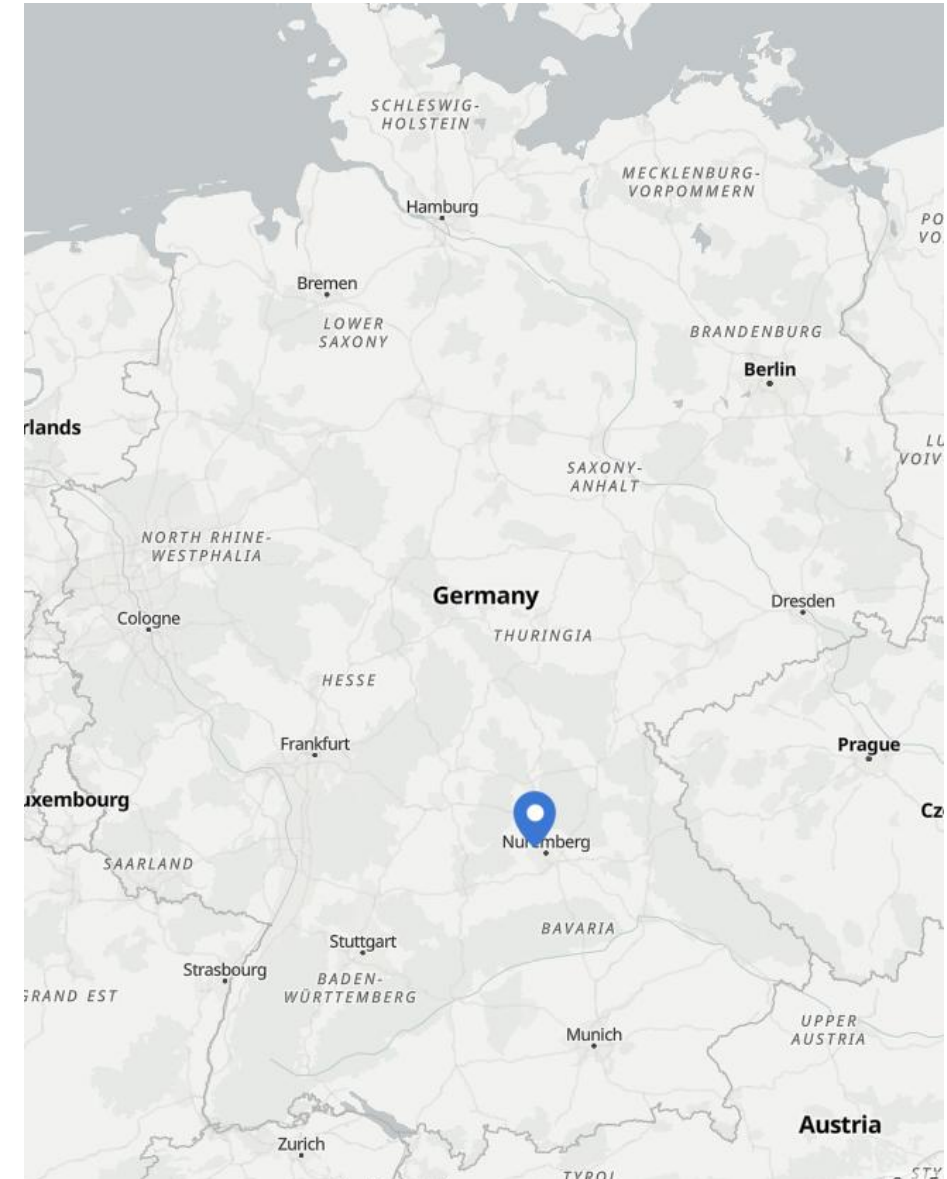
# Cities Showing Improving Affordability

■ ★ Fürth

Why:

- migration stabilised
- construction improved
- rental burden softened

Rare case where affordability improved → “emerging affordability market”.



# Executive Summary

1. Germany shows three structural affordability regimes —High Demand / Large Cities, Stable & Affordable Cities, Rising Pressure Cities.
2. 80% of cities remain stable, but 2020–2025 shows fastest transition on record.
3. 15 cities changed clusters; 12 of them moved *toward* high-pressure in the last 5 years.
4. Key affordability stress drivers are student density, migration, transactions, and supply bottlenecks — NOT prices.
5. Investment priority zones: Leipzig, Cologne, Dresden, Stuttgart, Nuremberg, Bremen.
6. Stable acquisition zones: Bamberg, Flensburg, Weimar.
7. Fürth is the *only* city showing improving affordability, a rare reversal.

# Cities That Changed Cluster

A total of **15 German cities** changed affordability cluster groups over 2010–2024; signaling structural pressure shifts.

The movement patterns are highly asymmetric:

## **Increasing pressure**

Bremen, Cologne, Dresden, Düsseldorf, Nuremberg, Stuttgart

## **Mid to high pressure**

Halle, Hamm, Herne, Jena, Kaiserslautern, Mulheim, Solingen, Pforzheim

## **Moderating pressure**

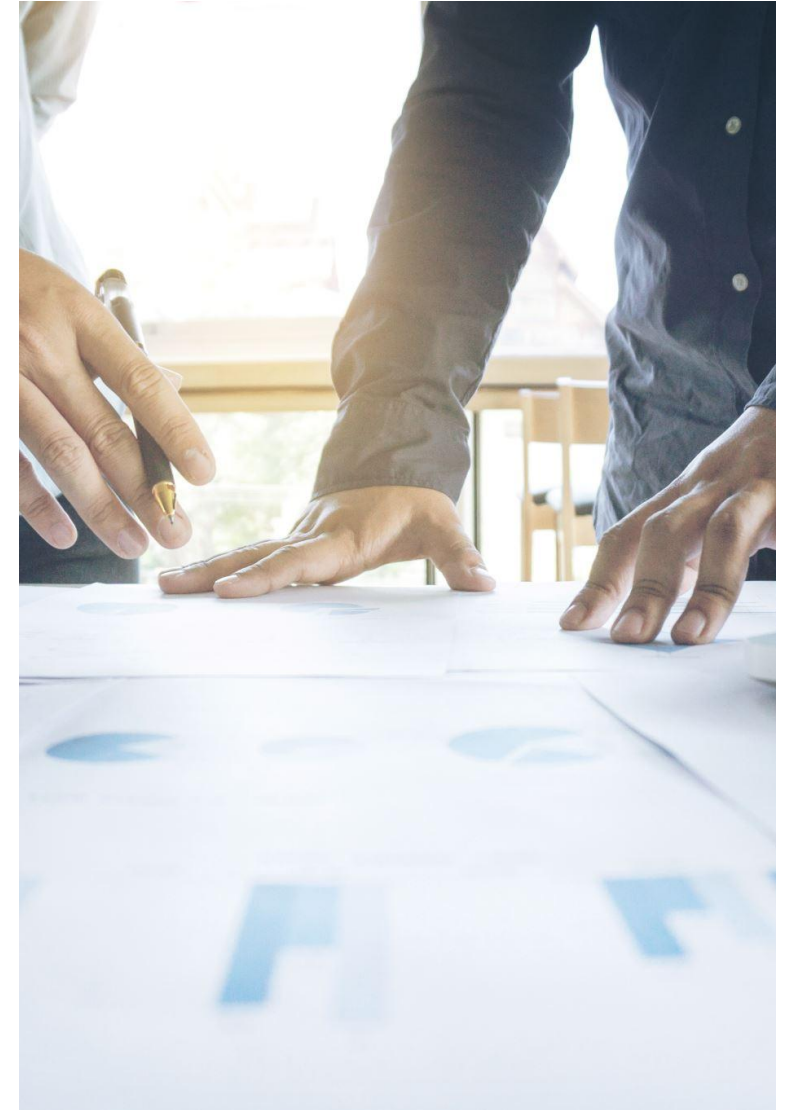
Fürth (only case)

*Most transitions (over 70%) occurred **after 2020**, showing accelerating affordability stress in Germany's major cities driven by: migration inflow, under-supply, rising construction cost, strong rental market pressure, student population growth, urban labour market shifts*



# Final Insights & Strategic Implications

- German cities follow clear structural patterns.
- K-means reveals consistent, interpretable clusters that remain stable ~80% of the time.
- The main determinants of affordability stress are NOT prices. Top SHAP drivers prove that:
  - student activity
  - migration
  - supply permits
  - building stock
  - transaction intensity

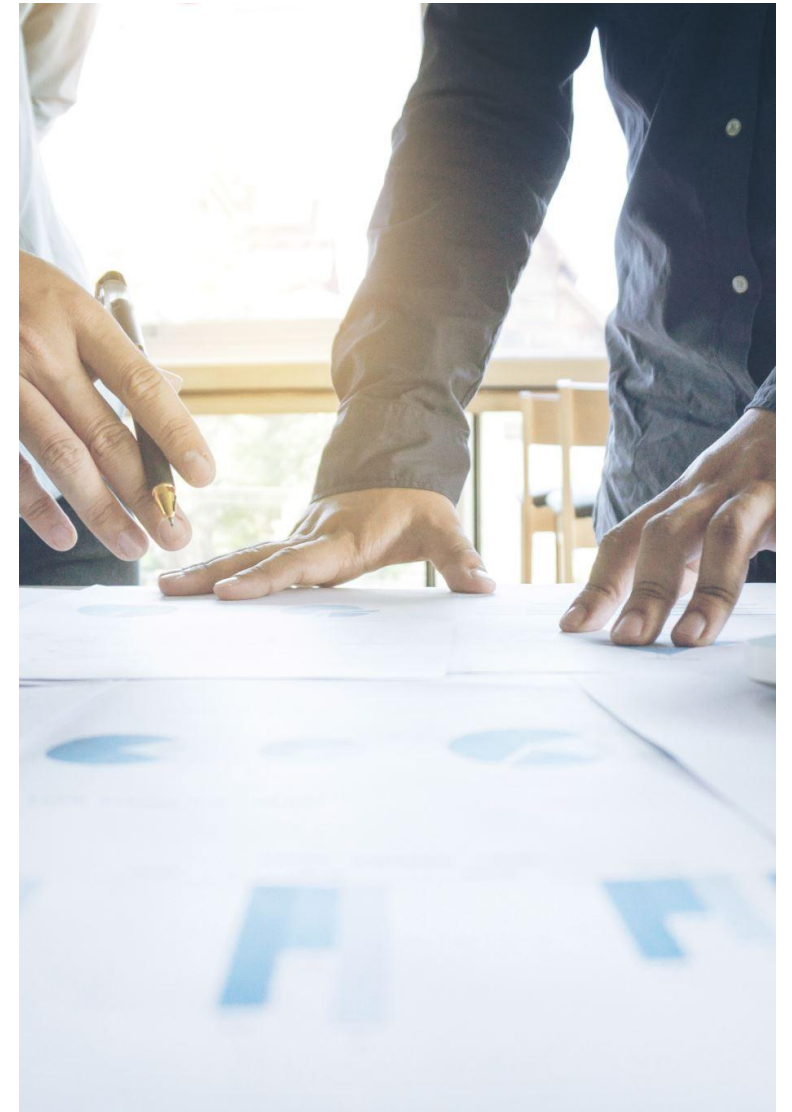


# Final Insights & Strategic Implications

- A set of cities is on a clear upward pressure trajectory.

Bremen, Nuremberg, Leipzig, Dresden, Cologne, Düsseldorf, Stuttgart

- ❖ These are *transforming into larger metropolitan magnets*.
- ❖ Stable cities remain reliably affordable.
- Cities maintain minimal changes even during market shocks.



# An attempt to validate

|                    |                   |                                  |                                                                                               |                                                                                                                  |
|--------------------|-------------------|----------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Secondary Drift    | Herne / Hamm      | Classified as "Yield Locations"  |  Confirmed | LEG Immobilien Report classifies these as high-yield/low-growth, confirming structural stagnation.               |
| (Demographic Drag) | Rural/C-Cities    | Real Price Drift: -1.0% to +1.0% |  Confirmed | Empirica Regio shows these regions lag behind A-city volatility; prices are "sticky" but falling in real terms.  |
| Structural Pattern | Nationwide        | 85% of districts stable          |  Confirmed | DZ HYP 2025 Report confirms "low volatility" outside Top 7; clusters are structurally fixed by building density. |
| Rent Pressure      | University Cities | Student Apts: Inv. Vol. ↑        |  Confirmed | Cushman & Wakefield reports surging investment in student housing due to severe supply shortages.                |

# An attempt to validate

| Cluster Type                 | Representative Cities          | Trend                                                                                                       | Validation Source           |
|------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------|
| 1. The "Industrial Drifters" | Wolfsburg, Solingen, Pforzheim |  Deepening Affordability | VW Crisis Reports (2024/25) |
|                              |                                | (Prices falling due to job insecurity)                                                                      | NRW.BANK Housing Report     |
| 2. The "Yield Kings"         | Herne, Hamm, Kaiserslautern    |  Sideways Drift          | LEG Immobilien Report       |
|                              |                                | (Stable rents, flat prices, high yield)                                                                     | CBRE-Empirica Vacancy Index |
| 3. The "Hidden Champions"    | Jena, Leipzig                  |  Defying the Drift     | JenaWirtschaft Report 2025  |
|                              |                                | (Rising rents, tech/student driven)                                                                         | JLL Market Overview         |

# An attempt to validate

3. Large expensive metros became more affordable (2020–2025)

Partially Confirmed

There is evidence of slight affordability improvements in several major markets since 2020, driven by cooling prices and income gains. Notably, **housing purchase prices fell** from their peaks in cities like Düsseldorf, Stuttgart, Frankfurt and Munich (declines of ~13–18% by 2024) [cushmanwakefie...](#), easing the ownership cost burden. At the same time, **rent growth slowed** in some high-cost metros. Industry data show that **Cologne and Düsseldorf maintained relatively stable rent-to-income ratios**, and cities like Dresden, Dortmund and Essen even saw *mathematical improvements* in affordability in recent years [realestate.bnppa...](#). However, this trend isn't universal – e.g. Berlin and Leipzig did not become more affordable, as rents there outpaced incomes [realestate.bnppa...](#). Overall, the pattern of slight relief in many previously “expensive” metros between 2020–2025 is **partially confirmed** by market reports.

# An attempt to validate

On the **rental side**, the trend was more mixed but still showed relief in certain costly markets. Rental growth in 2023–2024 slowed markedly in cities like Berlin and Munich, and even **declined slightly** for a period cushmanwakefie.... By 2024, Berlin's median rent was almost flat year-on-year (+1% jll.com), indicating that incomes (which kept rising) began to catch up with rents. The **rent freeze and subsequent regulation in Berlin** likely contributed to this effect. Meanwhile, Cologne and Düsseldorf – cited by the user – saw **rent increases roughly on par with income growth**, leaving their affordability relatively unchanged over the early 2020s. In fact, a report by BNP Paribas Real Estate finds **no deterioration in rent affordability in Cologne and Düsseldorf from 2015 to 2025**, labeling them as stable markets in terms of rent-to-income ratio realestate.bnppa.... **It even notes that Dresden and Dortmund became slightly more affordable (their average households now spend a lower percentage of income on rent than they did a decade ago)** realestate.bnppa.... Essen was also stable. These are precisely some of the cities the user listed.

# An attempt to validate

|                                                                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4. Secondary cities<br>(Herne, Hamm,<br>Sulingen, Pforzheim,<br>etc.) moved toward<br>higher affordability | Partially<br>Confirmed | <p>Many smaller and secondary cities remain highly affordable, and some have seen affordability improve gradually due to modest demand growth. For example, in the Ruhr area (Herne, Duisburg, etc.), typical housing costs take under 20% of a family's income, one of the lowest burdens nationwide <a href="#">immowelt.de</a> <a href="#">immowelt.de</a> . A 2025 housing atlas shows that in cities like Herne and Kaiserslautern, the cost of owning a home is now nearly on par with renting – a sign of very high affordability (mortgage payments require only ~0.1–0.5% more of income than rent) <a href="#">haufe.de</a> .</p> <p>These trends indicate that many secondary markets (including several listed by the user) have become more affordable or remained in the most affordable tier over 15 years. However, this is not true for every mid-sized city – e.g. Jena's housing has grown expensive for locals (families there face ~35% rent burdens, among the highest) <a href="#">immowelt.de</a> . Thus, the overall insight is valid for most named secondary cities, with a few exceptions.</p> |
|------------------------------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Jena is an interesting case were our studies show something but the Interwebs suggests something else 🤔

## Secondary Cities and Rising Affordability (Insight 4)

The user highlighted a set of secondary cities (Herne, Hamm, Solingen, Pforzheim, Kaiserslautern, Jena, Halle, Wolfsburg) that **gradually became more affordable**. For several of these, external data agrees that housing is very affordable, though not necessarily that it *improved* from 2010 – rather, it was always affordable or only mildly strained.

- **Ruhr Area cities (Herne, Hamm, etc.):** The Ruhr region has long had lower housing demand due to industrial decline and population stagnation. As a result, cities like **Herne and Duisburg consistently rank among the most affordable housing markets in Germany**. An immowelt analysis in 2021 found that in **Herne**, a family with a typical dual income spends **under 20% of net income on rent for a standard 4-room flat – versus well over 30% in Munich or Berlin**. Even lower-income households (no vocational qualification) face rent burdens in Herne below 25%, which is exceptionally low. This indicates that **housing costs in Herne (and similar Ruhr cities like Hamm or Gelsenkirchen) have remained very low relative to local incomes**, and indeed any change over 15 years has been minimal or towards even better affordability. Another metric: the Postbank Wohnatlas (2025) shows that **homebuyers in Herne need virtually the same income share as renters** for housing – a difference of only 0.1 percentage points. In other words, both renting and buying are comparably cheap. Such figures confirm that Herne is in the highest-affordability tier, and by implication, it hasn't left that tier in the last decade (if anything, economic challenges kept prices subdued).

- **Smaller cities in other regions (Kaiserslautern, Wolfsburg, etc.):** Kaiserslautern (a secondary city in Rhineland-Palatinate) similarly exhibits high affordability. The Wohnatlas data lists Kaiserslautern among cities where the **cost of owning a 70m<sup>2</sup> apartment is only ~0.4% of income more than renting** – effectively negligible. This suggests that housing is easily within reach for the average household. It's likely that since 2010, Kaiserslautern's modest growth and ample housing kept affordability steady or improved (especially if incomes rose due to its employment base like the US military and tech firms). **Wolfsburg is an interesting case: as the home of Volkswagen, it boasts very high wages which make housing costs (which are moderate) quite affordable**. While we didn't find a specific stat for Wolfsburg in the sources, it generally ranks as an affordable city to live in because of the wage-price mismatch in favor of incomes.
- **Solingen and Pforzheim:** These mid-sized cities (Solingen in North Rhine-Westphalia, Pforzheim in Baden-Württemberg) are somewhat different. Solingen, near Düsseldorf, has relatively low rents compared to the adjacent big cities, and a static population. It likely maintained decent affordability; for instance, Solingen's average rents are considerably lower than Cologne/Düsseldorf. Pforzheim, historically an industrial town, saw some house price increases last decade (benefiting from being between Stuttgart and Karlsruhe), but it remains cheaper than those hubs. We did not find direct evidence of improved affordability for these two, but by virtue of being outside the hottest markets, they are in more affordable ranges. Any changes from 2010 to 2025 would have been gradual.

- **Eastern secondary cities (Jena, Halle):** Here the evidence diverges. **Jena** was flagged by the user as moving toward higher affordability, but reports suggest Jena is actually *one of the less affordable* secondary cities due to its strong economy (university and tech sector) and limited housing supply. By 2021, Jena had among the highest rent burdens for families (around 35% of income for a typical family rental) [immowelt.de](#), placing it in the top 10 nationally in terms of rent strain. That implies Jena's affordability **worsened** over the past decade (as it used to be cheaper). **Halle (Saale)**, a larger city in East Germany, likely fared better: it has had only moderate rent growth and still relatively low prices. Halle's population has been stable or slightly rising, and wages inched up, so its affordability probably improved a bit or stayed comfortable.

In summary, **most of the cited secondary cities indeed have high or improving affordability**. Especially in the Ruhr and other economically weaker areas, housing cost increases were negligible compared to income growth, resulting in very low rent-to-income ratios [immowelt.de](#). Many households in these regions spend far below the 30%-of-income benchmark on housing. The **Postbank 2025 analysis confirms an abundance of "cheap" housing regions**: 165 regions where renters spend <13% of income on rent, predominantly outside the major metros [haufe.de](#). Cities like Herne, Kaiserslautern, Chemnitz, etc., fall in this category of enduring affordability. This validates the user's point, with the caveat that a few dynamic smaller cities (like Jena) did not follow the trend.

# Thank You



# Appendix Section



Cities that moved into a *better* affordability cluster (toward relief → cluster 1 or 0)

| City    | Original Cluster → Final | Net Direction                                                                                | Interpretation                                             |
|---------|--------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Fürth   | 2 → 1                    |  Improved | Only city with clear stress relief (supply ↑, migration ↓) |
| Bamberg | 1 → 2 → 1                |  Mixed    | Temporarily worsened, then recovered                       |
| Jena    | 1 → 2 → 1 → 2            |  Volatile | Mixed, unstable affordability                              |

# Cities showing multiple flips (unstable markets)

| City     | Flip Pattern       | Interpretation                     |
|----------|--------------------|------------------------------------|
| Bremen   | 0 ↔ 2 (four times) | High volatility in price vs supply |
| Essen    | 2 ↔ 0 ↔ 2          | Mixed signals (transaction-driven) |
| Mannheim | 2 ↔ 0 ↔ 2          | Labour/student driven              |
| Jena     | 1 ↔ 2 ↔ 1 ↔ 2      | Academic city volatility           |
| Münster  | 2 ↔ 0 ↔ 2          | Student inflow cycles              |

# Stable Markets



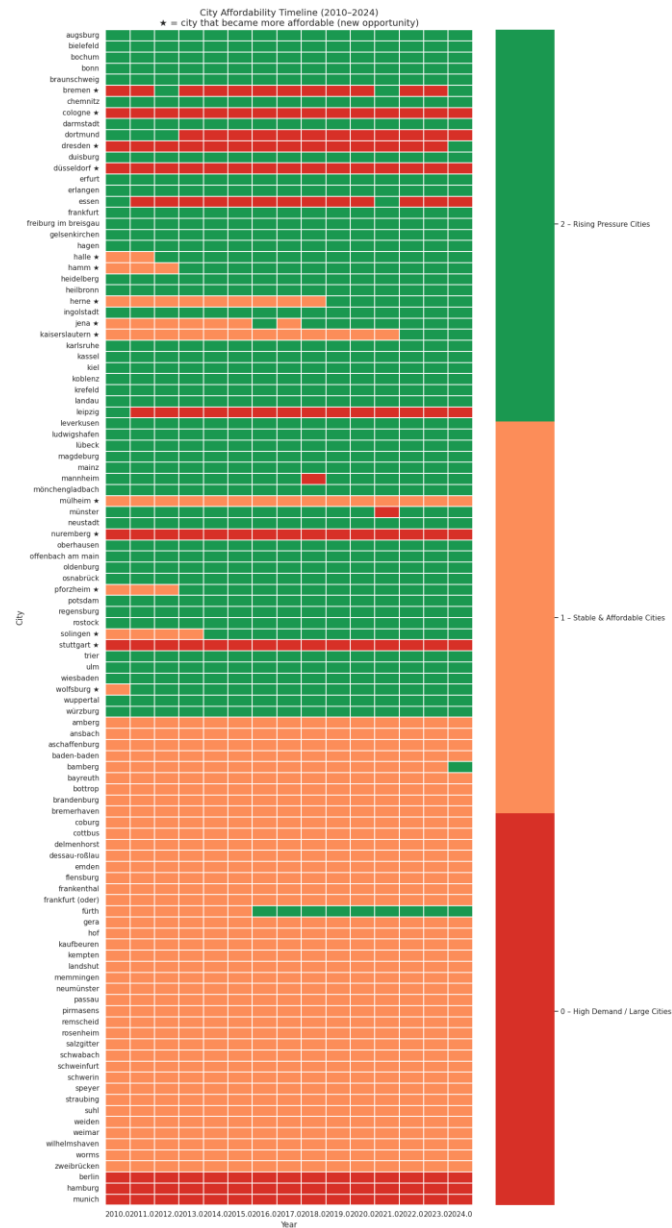
| Stable in Cluster 0   | Stable in Cluster 1 | Stable in Cluster 2        |
|-----------------------|---------------------|----------------------------|
| Berlin                | Amberg              | Augsburg                   |
| Hamburg               | Ansbach             | Bielefeld                  |
| Munich                | Aschaffenburg       | Bochum                     |
| Stuttgart (till 2024) | Bamberg (mostly)    | Bonn                       |
| ...                   | Bottrop             | Chemnitz                   |
| ...                   | Brandenburg         | Darmstadt                  |
| ...                   | Delmenhorst         | Dresden (till 2023)        |
| ...                   | Emden               | Duisburg                   |
| ...                   | Flensburg           | Erfurt                     |
| ...                   | Frankenthal         | Erlangen                   |
| ...                   | Frankfurt (Oder)    | Freiburg                   |
| ...                   | Gera                | Gelsenkirchen              |
| ...                   | Hof                 | Halle (after 2012)         |
| ...                   | Kaufbeuren          | Hagen                      |
| ...                   | Kempten             | Heidelberg                 |
| ...                   | Landshut            | Heilbronn                  |
| ...                   | ...                 | Kiel                       |
| ...                   | ...                 | Koblenz                    |
| ...                   | ...                 | Krefeld                    |
| ...                   | ...                 | Landau                     |
| ...                   | ...                 | Ludwigshafen               |
| ...                   | ...                 | Magdeburg                  |
| ...                   | ...                 | Mainz                      |
| ...                   | ...                 | Mannheim (after reversion) |
| ...                   | ...                 | Mönchengladbach            |
| ...                   | ...                 | Neumünster                 |
| ...                   | ...                 | Neustadt                   |
| ...                   | ...                 | Oldenburg                  |
| ...                   | ...                 | Osnabrück                  |
| ...                   | ...                 | Regensburg                 |
| ...                   | ...                 | Oberhausen                 |
| ...                   | ...                 | Offenbach                  |
| ...                   | ...                 | Potsdam                    |
| ...                   | ...                 | Rostock                    |
| ...                   | ...                 | Wolfsburg (before flip)    |

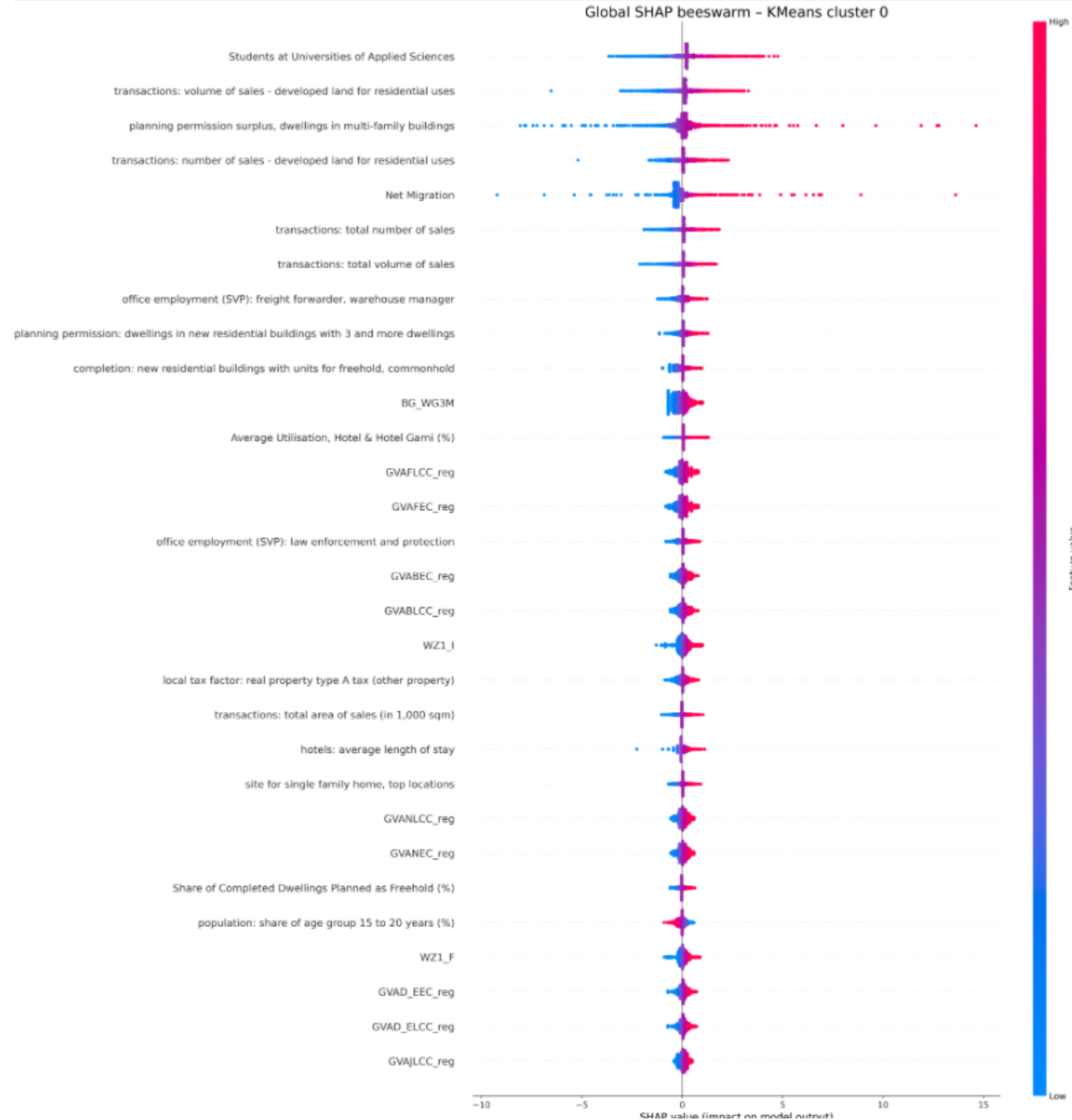
# SHAP Study

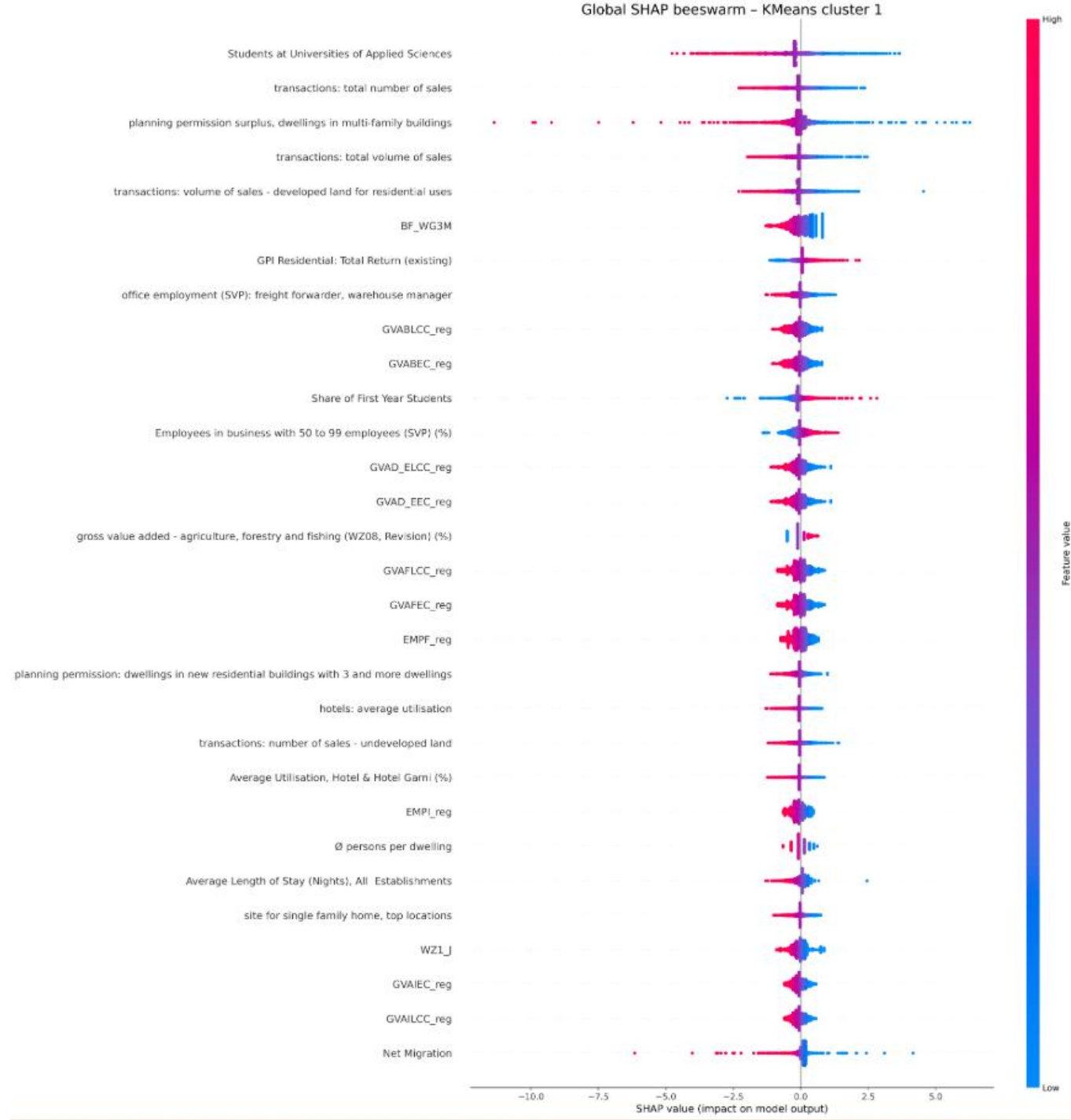
Cluster 2 has high SHAP contributions from income, transaction intensity, student presence, net migration, construction pipeline, GVA.  
Cluster 1 has SHAP contributions from lower income, unemployment, weaker migration.  
Cluster 0 has SHAP contributions from macro–metro characteristics but without extreme price pressure

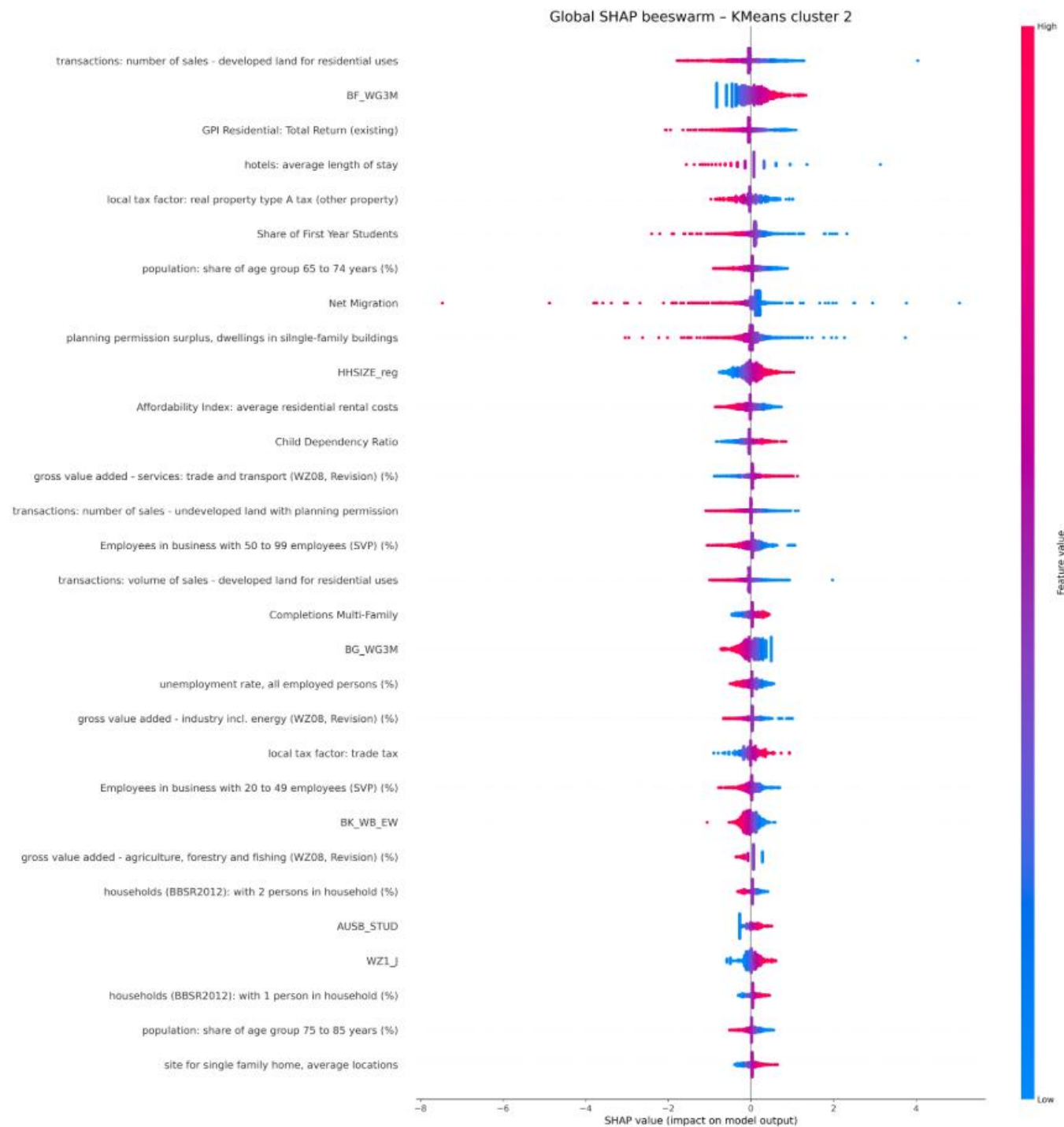
| Feature                                                      | What It Measures                                                                               | Why It Matters for Affordability                                                                                                    | Typical Effect on Cluster                                                                           |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Students at Universities                                     | Number or share of students enrolled in universities                                           | High student concentration → pressure on rental housing, short-term demand spikes, strong 1–2 room market.                          | ↑ Drives a city toward <b>Cluster 0–1</b> (higher pressure, less affordable).                       |
| Transactions: Volume of Sales – Developed Land (Residential) | Total euro volume of land transactions for residential use.                                    | High transaction volume signals active investor markets, speculative activity, and strong asset demand → upward pressure on prices. | ↑ Pushes cities into <b>Cluster 0</b> (expensive / premium).                                        |
| Planning Permission Surplus – Multi-Family Buildings         | Net balance between new multi-family building permissions and completions (pipeline pressure). | Excess pipeline = future oversupply relief. Low pipeline → tight future stock and rising rents.                                     | Low surplus → <b>Cluster 0</b> (tight). High surplus → <b>Cluster 2</b> (more affordable).          |
| Transactions: Total Number of Sales                          | Number of real estate units transacted overall.                                                | Many transactions = strong liquidity, high demand, investor & owner-occupier interest.                                              | ↑ Moves toward <b>Cluster 0–1</b> .                                                                 |
| Transactions: Number of Sales – Developed Land (Residential) | Count of land deals specifically for residential purposes.                                     | Strong indicator of future construction, developer confidence.                                                                      | High → can point to <b>future affordability improvement</b> (Cluster 2), depending on supply.       |
| Transactions: Total Volume of Sales                          | Overall capital volume traded across all asset types.                                          | Higher volumes imply strong market turnover, investment attention, and rising valuation pressure.                                   | ↑ Drives <b>Cluster 0</b> .                                                                         |
| Net Migration                                                | Inflow minus outflow of residents.                                                             | Strong migration inflows indicate attractive cities → rents rise, supply tightens, affordability decreases.                         | Positive migration → <b>Cluster 0 or 1</b> . Negative → <b>Cluster 2</b> .                          |
| BF_WG3M (Building Footprint – Multi-unit buildings)          | Share of buildings that are multi-family or multi-unit structures.                             | Captures the density and intensity of urban dwelling stock. High MF density → tight rental markets and strong demand.               | High BF_WG3M → <b>Cluster 0</b> .                                                                   |
| GPI Residential: Total Return (existing)                     | Total return (income + capital) of existing residential properties (Global Property Index).    | High investment returns signal price momentum + rental yield pressure → strong investor appetite reduces affordability.             | High returns → <b>Cluster 0</b> .                                                                   |
| Office Employment (SVP): Freight, Warehouse Managers         | Employment in logistics, freight, warehousing, supply chain management.                        | Logistics-heavy cities grow fast due to e-commerce, requiring workers → increasing demand for rental housing.                       | Strong employment → <b>Cluster 1</b> (mid-pressure) or <b>Cluster 0</b> if combined with migration. |

# Complete Heatmap. Enlarge to Study.









Examples:

► Stuttgart (0→2)

rising GVA, strong employment, strong housing price growth

SHAP: strong push toward cluster 2 from GVA, transactions, prices, migration

Matches perfectly.

► Cologne (0→2)

rising students, transactions, migration, price level

SHAP: those are top-5 SHAP drivers

So this city behaves exactly like the SHAP profile of cluster 2.

► Halle / Jena (1→2)

increases in student share, construction, migration, employment density

SHAP: those are top drivers pulling toward cluster 2